# Thilak D

Data Scientist and Chief of Staff

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## SUMMARY

I'm a data scientist working as Chief of Staff at Mondee on our healthcare AI product. I built the evaluation platform we run every release, analysed 50,000+ hospital encounters, and coordinate our pilots in six hospitals. I also work on the CDSCO filing and business planning with leadership. MSc in Data Science and Computing; doing an MBA at BITS Pilani.

## EXPERIENCE

**Mondee Tech Private Limited (Mondee Tabhi Group)** — Hyderabad

*Chief of Staff (Data Science and AI)* — Oct 2025 – Present

Report to the Group Chief Growth Officer and work across our healthcare AI product.

- Built our evaluation platform with the data science team. It checks every AI summary against the doctor's edited version, scores ASR and LLM output, and runs each release so the product team can review AI and engineering work.
- Fine-tuned Google's MedASR on 15,000 Telugu consultation recordings. The vendor we already used was still more accurate (7.5% vs 10.5% word error rate, 2.8% vs 4.2% on medical terms) and faster, so we stayed with it. Pipelines for Hindi, Tamil and Bengali are ready.
- Analysed 50,000+ patient encounters and 300+ doctor records and found that paused consultations were later marked finished with no record of whether they resumed. Worked with the dev team on the fix, and that error is now close to zero.
- Coordinate our pilots across six hospitals, including government hospitals. In one pilot, 4 doctors reviewed about 3,000 records over a month and accepted 88% of AI summaries without edits. Their main feedback was that sorting records took too long.
- Built a React and FastAPI dashboard on doctor review data that all six pilot hospitals now use. It tracks KPIs and sends flagged fields to the AI, product or dev team to fix.
- Prepared our CDSCO application for the product as a Class A medical device: post-market surveillance plan, PSUR, ISO 13485 internal audit and the MD-12 test licence, which is now under review.
- Lead a team of 6 that tests the patient and doctor apps (Android, iOS, web) before every release and follows up on fixes with the dev and AI teams. Also keep the release tracker leadership uses; we ran 3 release cycles over the past year.
- Built the capex, opex and 3-year P&L model for a planned USD 100M care network: 4 inpatient hubs, 80 diagnostic centres, and mobile doctor and nurse units. The plan is now being implemented.

**Sentinel AI ML Labs (SAIMA Labs)** — Chennai (part-time)

*Lead, AI ML Special Projects (AI Data Science Analyst)* — May 2025 – Nov 2025

*Junior Data Scientist* — Jun 2024 – May 2025

- Built a dental health prediction model and automated WhatsApp messages to patients for a dental hospital group.
- Built a risk scoring bot (RAS) that rates startups for an investment foundation, using 20+ of its datasets. It works for one startup or a bulk CSV.
- Worked with cardiologists on an AI cardiac risk prediction model for a MedTech startup.
- Mentored 5 interns across the RAS bot, product and website projects.

**TaskBench** — Chennai (internship)

*Data Analyst (internship)* — May 2024 – Jun 2024

- Built prediction and classification models in Python for market research, improving efficiency by 50%.
- Analysed 50,000+ records to give clients input for their marketing plans.
- Prepared 2 due diligence reports and 3+ client proposals.

## RESEARCH AND PROJECTS

**SAGE (Semantic and Graph-based Engine): Hybrid Knowledge Retrieval for Ayushman Bharat Treatment Guidelines**

*Conference paper. Thilak D (first author), P Sunil Kumar, SVSN Sarma. Presented at the 13th International Conference on Contemporary Engineering and Technology (ICCET 2025), Chennai, Mar 2025. Paper ID ICCET250837.*

- Fine-tuned LLaMA 3.1 (8B) and Phi-4 to turn plain questions into graph queries (Cypher). Phi-4 did better, with 0.86 BLEU and 0.73 exact match.
- Combined graph search with vector search. Answer relevance went from 0.89 with plain RAG to 0.95, and faithfulness from 0.87 to 0.94.

**Hybrid question answering with LangGraph: multi-hop RAG and knowledge graphs, MSc thesis, SSSIHL** Jun 2024 – Apr 2025

- Extended the paper into the MSc thesis: a LangGraph agent that answers Ayushman Bharat questions, like which private hospitals in a state do a given surgery and what documents to bring.
- Collected 150+ scheme documents using RPA and built a knowledge graph of 50,000+ hospitals, 1,600+ surgeries and 9 states.

**Automated job recommendation system, SSSIHL** Nov 2023 – Apr 2024

- Pulled required skills out of job descriptions with Sentence Transformers, which improved profile matching accuracy by 25%.

## EDUCATION

**MBA, Strategy and Consulting** Expected 2028

BITS Pilani, Work Integrated Learning Programme (for working professionals)

**MSc, Data Science and Computing** Jul 2023 – May 2025

Sri Sathya Sai Institute of Higher Learning (SSSIHL). CGPA 8.3. Thesis: hybrid question answering with LangGraph, multi-hop RAG and knowledge graphs.

**BSc (Hons), Computer Science** Jun 2020 – Jun 2023

Sri Sathya Sai Institute of Higher Learning (SSSIHL). CGPA 7.1

## SKILLS

**AI and ML:** Machine learning, Deep learning, NLP, Predictive modelling, Speech recognition (ASR), Explainable AI (SHAP), Computer vision

**LLMs and generative AI:** RAG, Agentic AI (LangGraph), LangChain, Knowledge graphs, Ontology design, Hybrid vector and graph retrieval, LLM fine-tuning (Unsloth), Text-to-Cypher, LLM evaluation (BLEU, ROUGE, faithfulness), Small task-specific models

**Programming:** Python, SQL, R, JavaScript, TypeScript, C/C++, Java

**Data and backend:** pandas, NumPy, scikit-learn, PyTorch, TensorFlow, PySpark, Neo4j (Cypher), Weaviate, MongoDB, MySQL, FastAPI, Flask, AWS

**Strategy and operations:** Financial modelling (capex, opex, P&L, IRR), Market-entry strategy, CDSCO / MDR 2017, ISO 13485, ABDM / ABHA, Release management, Pitch decks, Power BI, Advanced Excel

## ADDITIONAL INFORMATION

**Certifications:** Neo4j Certified Professional; AWS Academy Graduate: Cloud Foundations; Supervised Machine Learning: Regression and Classification; Databases and SQL for Data Science; Power BI

**Honours:** Best Intern 2024, Sentinel AI ML Labs; Ranked 15th nationally in Analytics Olympiad 2023, a machine learning hackathon hosted by MachineHack; Cleared the Actuarial Common Entrance Test (ACET), March 2023; House Championship Award, SSSIHL

**Conferences:** Presented paper, 13th International Conference on Contemporary Engineering and Technology (ICCET 2025), Chennai (Mar 2025); Presented and pitched, Global Innovation Summit and Retreat, Colombo, Sri Lanka (Sep 2025); Presented, ICE on I: PHILIPPiNEXT, 1st International Conference and Exposition on Innovations, Okada Manila, Parañaque City, Philippines (Jul 2025)

**Leadership:** Student President, Digital Enlightenment Club, SSSIHL (2023–2025); Co-founder, Eco-Einsteins Club, SSSIHL (2024–2025); Member, Institute Innovation Council, SSSIHL (2024); Captain, Annual Sports and Cultural Meet, SSSIHL (2025)

**Languages:** Tamil (native), English (fluent), Telugu (conversational)

**Interests:** Video editing, Music production, Reading tech blogs, Badminton, Cricket, Volleyball